State Diagram

What is State Diagram

- It presents the states an object can be in along with the transitions between the states and shows the starting point and endpoint of a sequence of state changes.

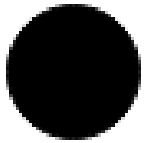
- Examples
    1. Switching light, its state changes from Off to On.
    2. Clicking a remote control, TV changes its state from showing you one channel to showing you another.
    3. After an appropriate amount of time, a washing machine changes its state from Washing to Rinsing.
    4. project: submitted, evaluated, accepted, rejected, monitored, audited, finished, delayed

2-Difference Between State Diagram and Other Diagrams

- The diagrams you've already studied model a **group of classes, objects, or use cases**.

- A **state diagram** shows the states of a **single object.**

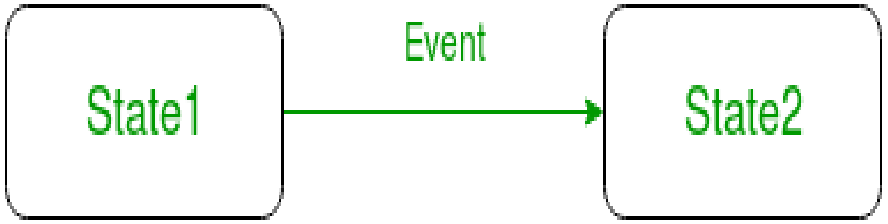
State diagram symbols



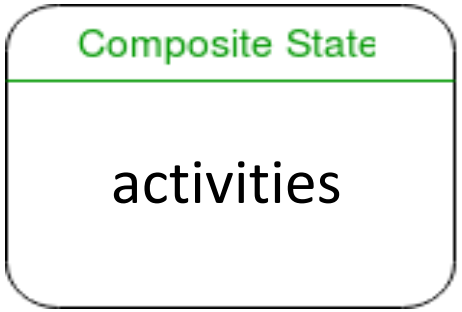
Start symbol



State symbol

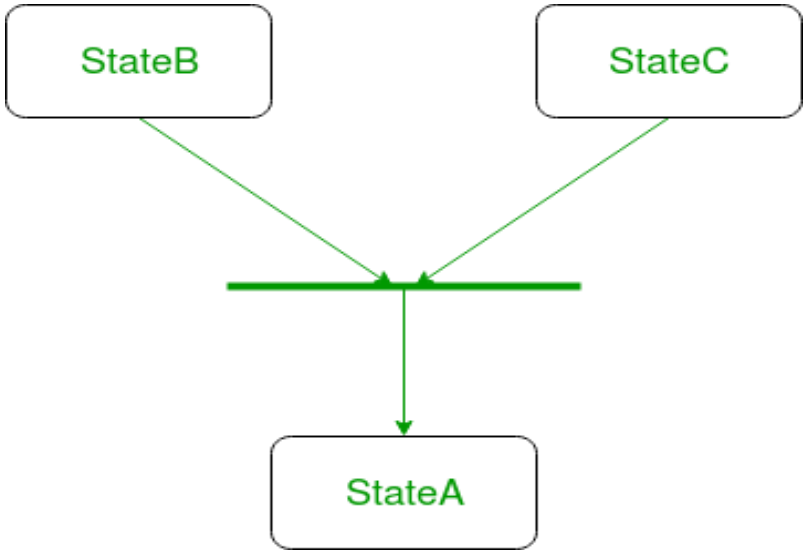


event symbol

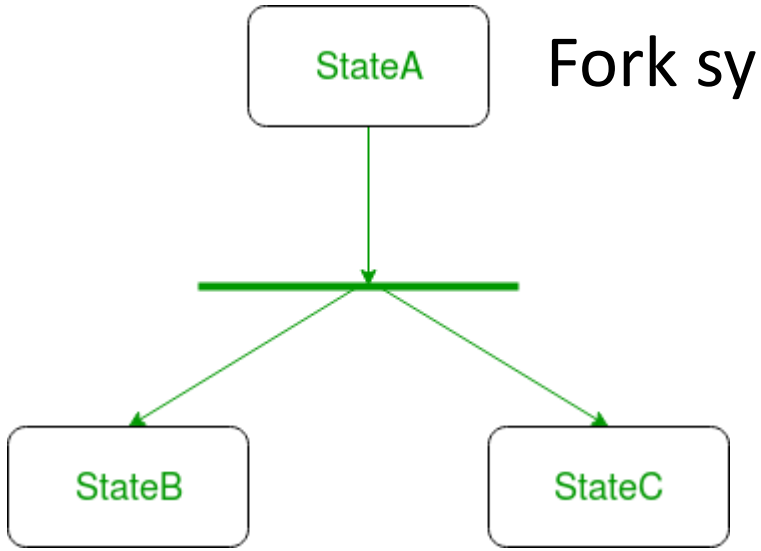


Composite State symbol

State diagram symbols



Join symbol



Fork symbol



Final symbol

State example

- **Example**

Considering the example shown in the above figure, the transition from Waiting state to Riding state takes place when the person gets a taxi. Likewise, the final state is reached, when he reaches the destination. These two occurrences can be termed as events Get_Taxi and Reach_Destination. The following figure shows the events in a state machine.



State machine Diagram example

Car Agency problem

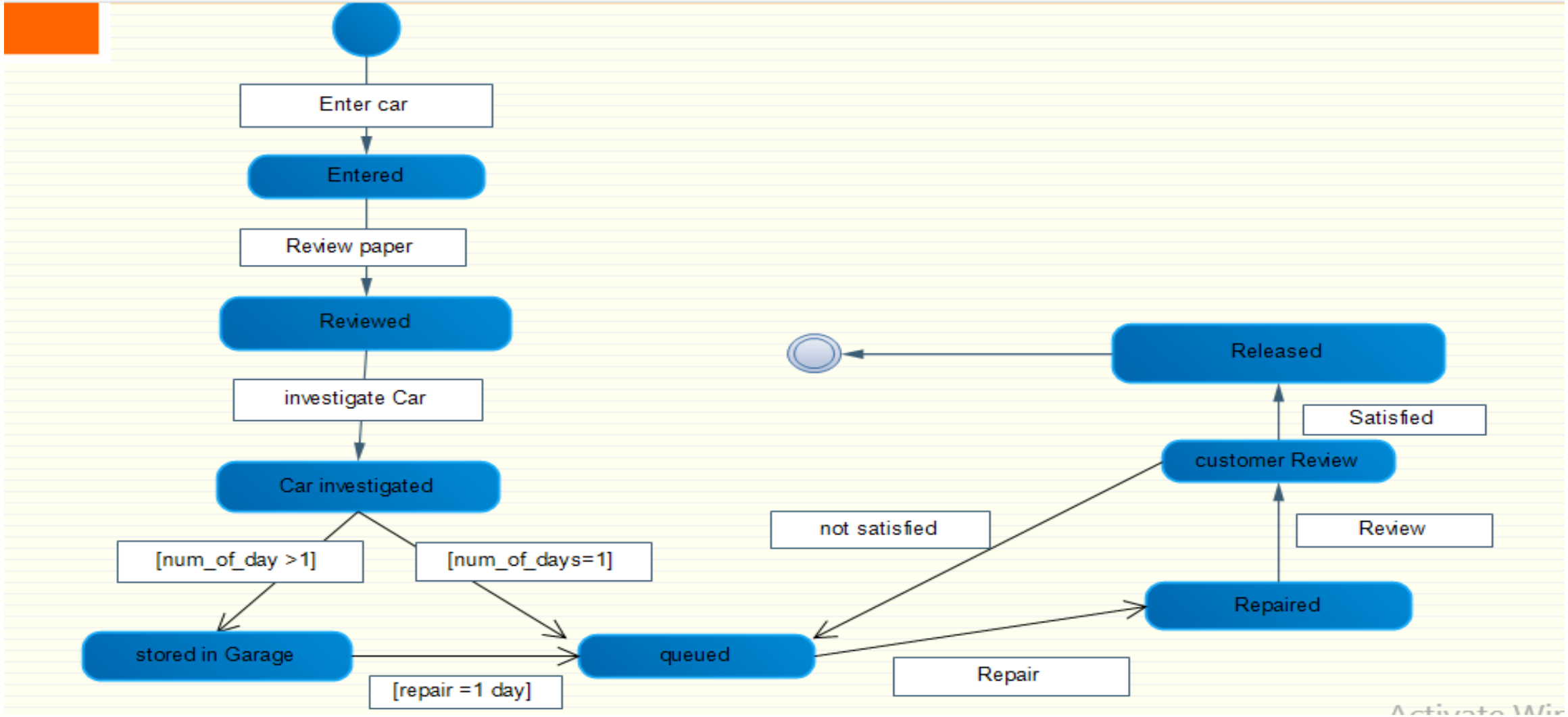
In car Agency : the customer enters the car then car’s papers reviewed after that the car is investigated if the repair take one day

So it will be queued, if it is more than one day it will be stored in Garage.

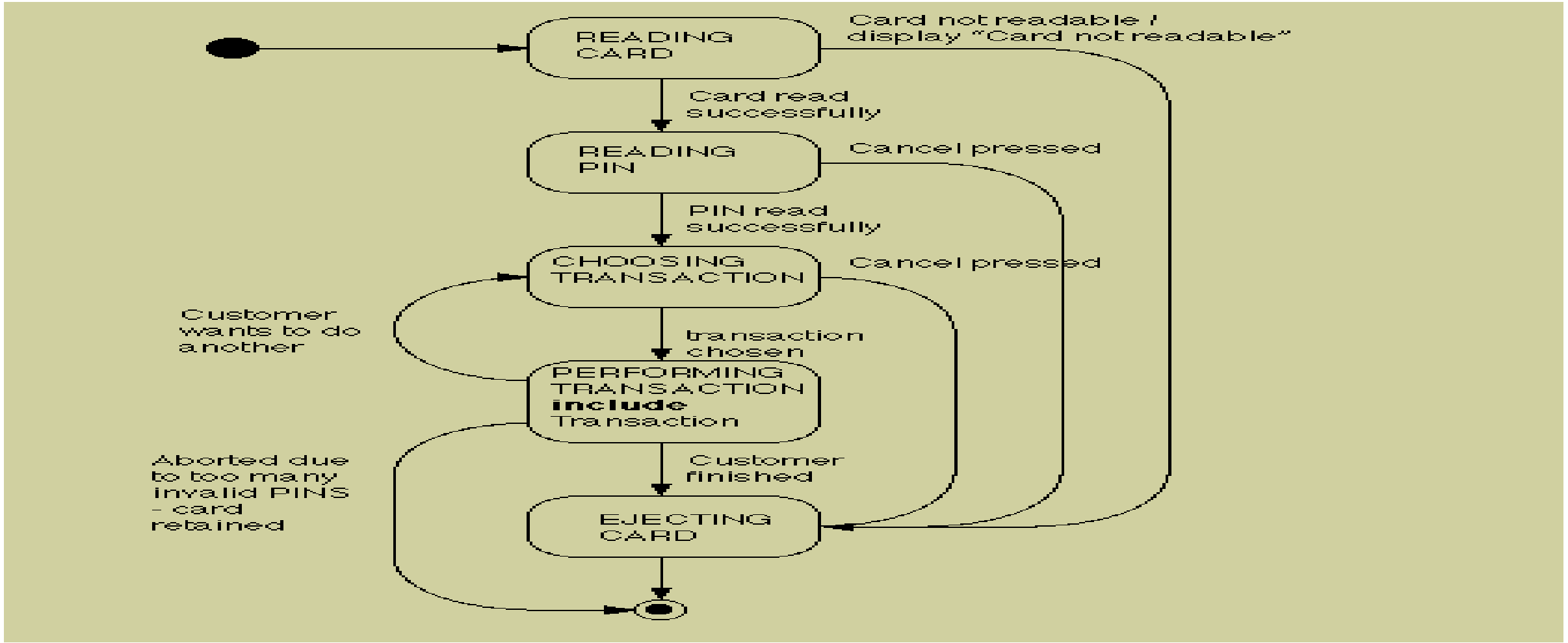
It will be moved from Garage to queue when repair day is today .

It will be moved from the queue to customer check after repaired ,if customer not satisfied it will be queued if satisfied it will be Released.

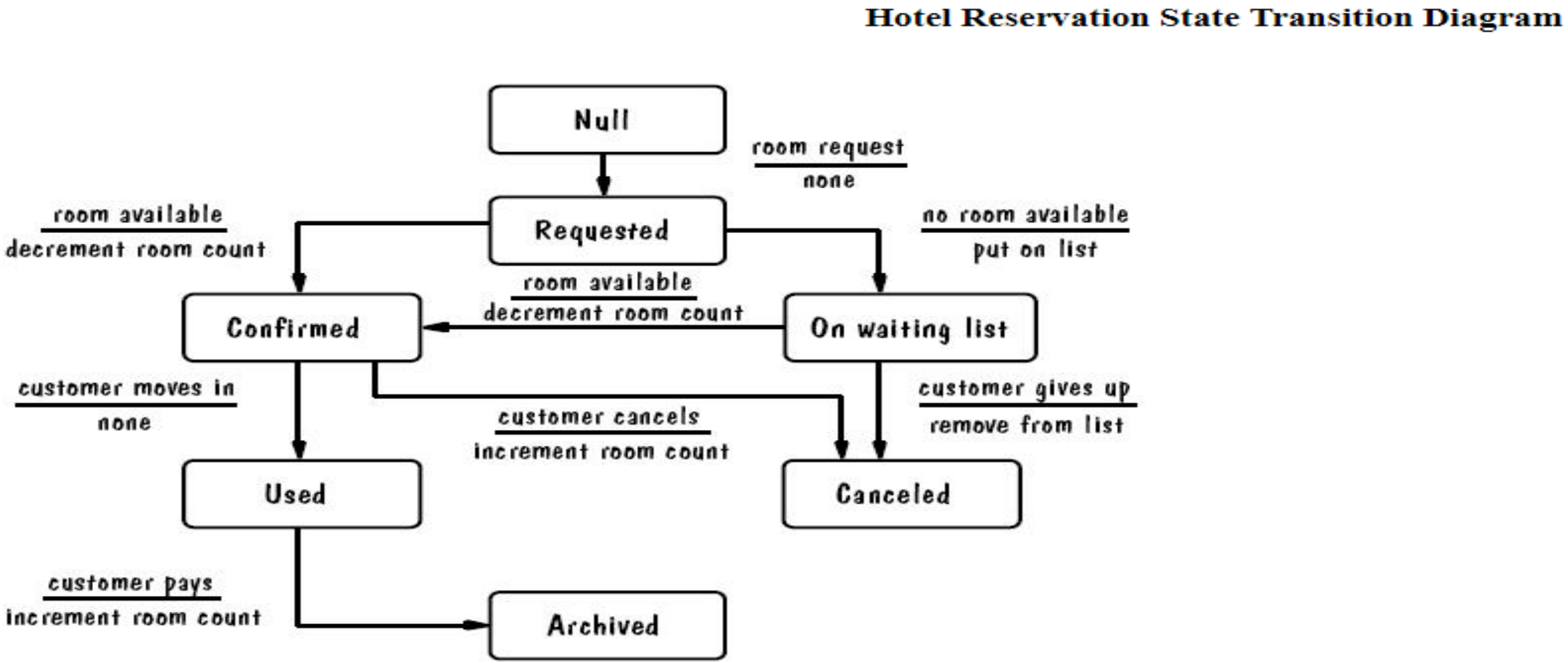
State machine Diagram for car agency



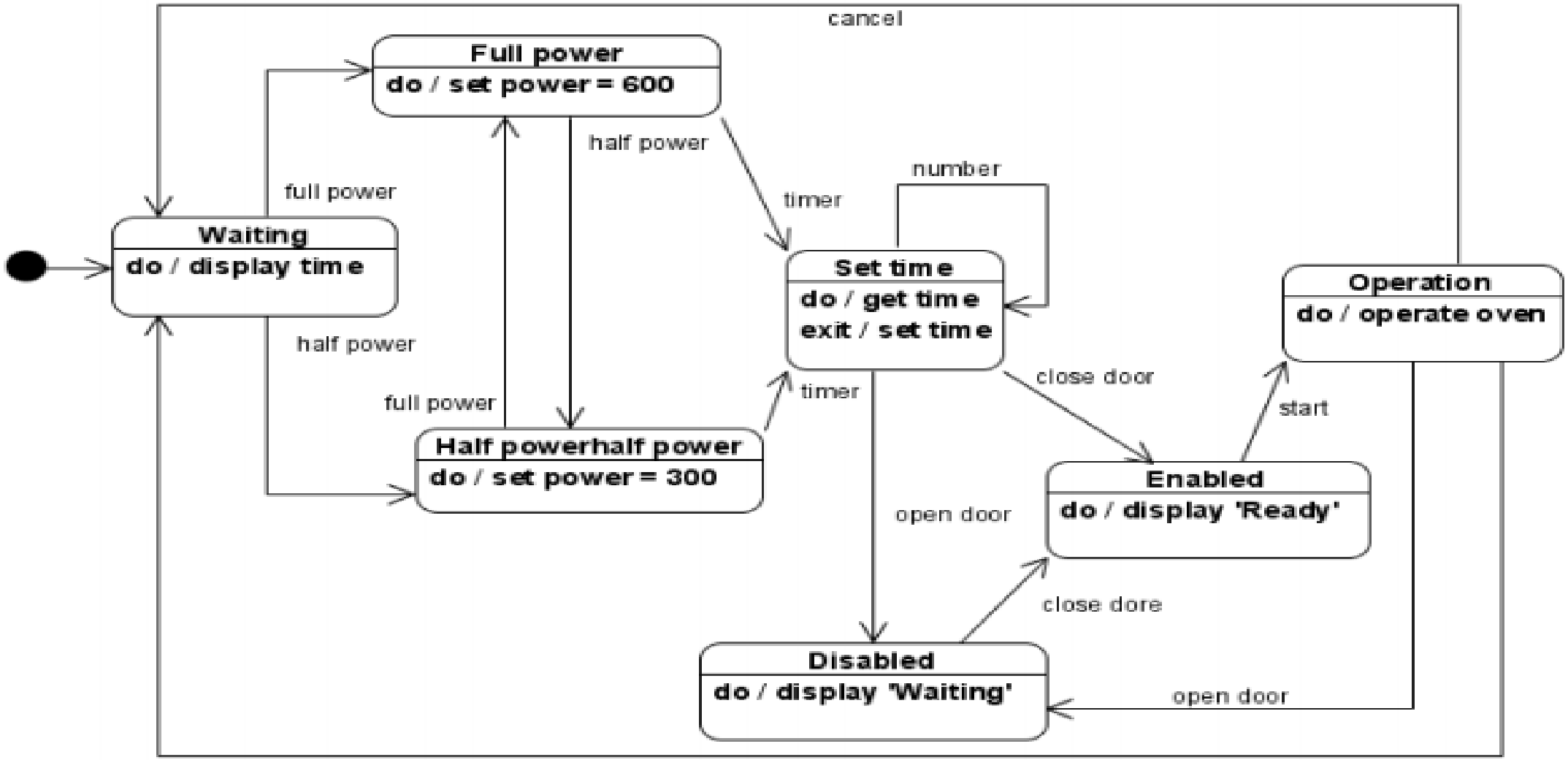
State Diagram for ATM



Hotel state machine diagram



State Diagram for microwave oven



Thank you